

PSET 5: Functions of several variables and optimization with several variables

Assignment Qs

- Name
- How long did this problem set take you (round to nearest hour)? 0-1 hour 2-3 hours 4-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Evaluating multivariate functions

Recall the spatial model of policy from class. Policy is N -dimensional, $\mathbf{x} \in \mathbb{R}^N$, and legislator i 's utility $U : \mathbb{R}^N \rightarrow \mathbb{R}^1$ is

$$U(\mathbf{x}) = - \sum_{j=1}^N (x_j - \mu_j)^2,$$

where $\boldsymbol{\mu} = (\mu_1, \dots, \mu_N)$ is the legislator's ideal point. Suppose $N = 3$ and $\boldsymbol{\mu}_1 = (0, 0, 0)$.

- (0.5 points) Evaluate $U(\mathbf{m})$ for the policy proposal $\mathbf{m}_1 = (1, 2, -1)$, and for the proposal $\mathbf{m}_2 = (0, 1, 0)$. Show your work.
- (0.5 points) Which proposal does the legislator prefer, and how do you know from the values you just computed?
- (0.5 points) (in words only) How could you identify the legislator's preferred policy computationally, if you hadn't been told $\boldsymbol{\mu}$ in advance. Be specific using terms and concepts from class.

Problem 2: First and second partial derivatives

Find the requested partial derivatives of each function.¹

- (0.5 points) All first partial derivatives of $f(x, y) = \frac{1}{3}x^5 - 3x^3y^2 + 3xy^4$
- (0.5 points) All first partial derivatives of $g(x, y) = xe^{4y}$
- (0.5 points) All first partial derivatives of $k(x, y) = \frac{x+y}{x-y}$

¹Inspired by Grimmer HW6.3

- d. (0.5 points) Returning to f from part (a), find the second partial derivatives $\frac{\partial^2 f}{\partial x^2}$ and $\frac{\partial^2 f}{\partial y^2}$, and both cross partials $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 f}{\partial y \partial x}$. What do you notice about the two cross partials?

Problem 3: Gradients, Hessians, and Jacobians

- a. $g(x, y) = x^4 - 3x^2y^3$
- (0.5 points) Find the gradient ∇g and evaluate it at $(x, y) = (1, 1)$.
 - (0.5 points) Find the Hessian $\mathbf{H}(g)$.
- b. $f(x, y) = x^3y^4 - 3xy + 1$
- (0.5 points) Find the gradient ∇f and evaluate it at $(x, y) = (0, 0)$.
 - (0.5 points) Find the Hessian $\mathbf{H}(f)$.
- c. (0.5 points) The gradient is the derivative of a *scalar-valued* function; the Jacobian is the matrix of derivatives of a *vector-valued* function. Consider $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$\mathbf{F}(x, y) = (x^2y, x + y^3).$$

Find the Jacobian $\mathbf{J}(\mathbf{F})$. Then, in one sentence, explain why it would not be meaningful to ask for the Jacobian of g in part (a) in the same sense.

Problem 4: Find the critical points

(1.5 points) Find the local minimum values, local maximum values, and saddle point(s) of

$$f(x, y) = x^3 + y^3 - 3xy.$$

Follow the process we discussed in class: calculate the gradient, set it equal to zero and solve the resulting system of equations, calculate the Hessian, and assess the Hessian at each critical value using the determinant test. Be sure to show your work on each of these steps, and state clearly what kind of point each critical value is.²

Problem 5: Definite and indefinite integrals

Solve the following definite or indefinite integrals using the antiderivative method.³

Reminder: the basic approach to compute the definite integral of $f(x)$ from a to b is by using the formula $F(b) - F(a)$, where $F(x)$ is the **antiderivative** of f .

- (0.5 points) $\int_{-1}^0 (3x^2 - 1) dx$
- (0.5 points) $\int_2^4 e^y dy$
- (0.5 points) $\int_3^3 \sqrt{x^5 + 2} dx$
- (0.5 points) $\int \left(x^2 - x^{-\frac{1}{2}} \right) dx$

²Inspired by Grimmer HW7.4

³Inspired by Gill 5.10, Grimmer HW4.1, Gill 5.13, and 5.14

Problem 6: Substitution and integration by parts

- a. (0.5 points) $\int 2x(x^2 + 1)^4 dx$
- b. (0.5 points) $\int \frac{3x^2}{x^3 + 5} dx$
- c. (0.5 points) $\int xe^{2x} dx$
- d. (0.5 points) $\int \log(x) dx$ (*Hint: you may take $dv = dx$.*)
- e. (0.5 points) Without solving, state whether you would use substitution or integration by parts for each, and why:
 - i. $\int 5x^4 (x^5 - 2)^3 dx$
 - ii. $\int x^2 \log(x) dx$

Problem 7: Improper and multiple integrals

For (a)–(c), determine whether the integral is convergent or divergent, and evaluate those that are convergent.⁴ Note that these are improper for different reasons — say which in each case. For (d)–(e), evaluate the iterated integral.

- a. (0.5 points) $\int_1^\infty \left(\frac{1}{3x}\right)^2 dx$
- b. (0.5 points) $\int_{-\infty}^0 x^3 dx$
- c. (0.5 points) $\int_0^1 x^{-\frac{1}{2}} dx$
- d. (0.5 points) $\int_0^1 \int_2^3 x^2 y^3 dx dy$. Does the order in which you integrate matter here? Explain why or why not.
- e. (0.5 points) $\int_0^1 \int_0^{\sqrt{1-x^2}} 2x^3 y dy dx$. State which variable you must integrate first and why — contrast your answer with part (d).

Problem 8: Interpretation

No new computation is required for this problem. Answer each part in two to four sentences of prose.

- a. (1 point) **Reading a partial derivative.** Suppose we regress presidential approval in month i on the employment rate and the price of gas, and obtain

$$\text{Approval}_i = 0.8 - 0.5 \text{Employ}_i - 0.25 \text{Gas}_i.$$

Write down $\frac{\partial \text{Approval}_i}{\partial \text{Employ}_i}$ and interpret it substantively. Your interpretation should make clear what is happening to Gas_i while employment changes, and why that matters for the claim you are making.

⁴Inspired by Grimmer HW4.3

- b. (1 point) **Gradient versus Hessian.** Explain why the condition $\nabla f(\mathbf{a}) = \mathbf{0}$ is *necessary* but not *sufficient* for $\mathbf{a}$ to be a local extremum. Use your answers to Problem 4 to illustrate: both $(0, 0)$ and $(1, 1)$ have zero gradient, but they are different kinds of point. What does the Hessian tell you at each that the gradient could not? Finally, explain what it means when the determinant test returns $AC - B^2 = 0$, and why we cannot classify the point in that case.
- c. (0.5 points) **Integrals and probability.** A continuous probability density f must satisfy $\int_{-\infty}^{\infty} f(x) dx = 1$. Explain what this requirement means, and what $\int_a^b f(x) dx$ represents for some interval $[a, b]$.

AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and *how* it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.